J-C-138b
September 20, 1962

SUPERSEDING
Int. Fed. Spec. J-C-00138a(GSA-FSS)
October 12, 1961 and
Fed. Spec. J-C-138
April 8, 1943

FEDERAL SPECIFICATION

CABLE AND WIRE, POWER, ELECTRICAL; VARNISHED-CLOTH-INSULATED, (0- to 5000-VOLT SERVICE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cable and wire employing solid or stranded soft-annealed copper conductors and varnished-cloth insulation for 0- to 5000-volt service.

1.2 Classification.

1.2.1. *Types, classes, styles, and sizes.* Cable and wire furnished under this specification shall be of one type and the following classes, styles, and sizes, as specified.

Classes:

Class 1.—Solid conductor.

Class 2.—Concentric-lay-stranded conductors of standard flexibility, conforming to class B of ASA Standard C7.8.

Class 3.—Concentric-lay-stranded conductors when greater flexibility is desired, conforming to class C of ASA Standard C7.8.

Class 4.—Rope-lay-stranded conductors when extreme flexibility is desired, conforming to class G of ASA Standard C7.13.

Styles:

Style V.—Single conductor having a fibrous covering.

Style VD.—Twin wire having two conductors laid parallel under an outer fibrous covering.

Style VM.—Cable having two or more conductors twisted together under an outer fibrous covering.

NOTE: The Letter "L" when used as a suffix in combination with any of the above designations indicates the application of a lead sheath.

Sizes:

AWG 14-2,000 MCM.—Single conductor.

AWG 14-1,000 MCM.—Multiple conductor.

1.2.2 *Service voltage.* The maximum working voltages of cable and wire in accordance with this specification shall be 600, 1000, 2000, 3000, or 5000 volts. The letter designations shown in 1.2.1 when used alone indicate conductors for use at not more than 600 volts. Cable and wire for use at higher voltages shall be indicated by adding numeral suffixes to the style letters in 1.2.1 as follows:

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10—for use at not more than 1000 volts.
20—for use at not more than 2000 volts.
30—for use at not more than 3000 volts.
50—for use at not more than 5000 volts.

For example, VM-L represents a multiple-conductor cable having a lead sheath and designed for use at not more than 600 volts, while VM-L30 represents a multiple-conductor cable having a lead sheath and designed for use at not more than 3000 volts.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

J-C-98—Cable and Wire, Insulated; Methods of Sampling and Testing.
UU-P-271—Paper, Wrapping, Waterproof, Kraft.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547—Liner, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Inspection Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications, of the issues in effect on date of invitation for bids, form a part of this specification:

Underwriters' Laboratories, Inc., Publication:

UL-133-Standard for Varnished-Cloth Wires and Cables.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Illinois; or 1655 Scott Blvd., Santa Clara, California.)

American Standards Association, Inc. Publications:

C7.8-1957 — Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft, Specification for.
C7.13-1959—Rope-Lay Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors, Specifications For.

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(Copies may be obtained from American Standards Association, Inc., 10 East 40th Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Cable and wire furnished under this specification shall conform to the requirements of Underwriters' Laboratories, Inc., Standard No. 133 for Varnished-Cloth Wires and Cables, and the applicable American Standards Association, Inc. Standard.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the wire and cable he proposes to supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc., Standard No. 133 for Varnished-Cloth Wires and Cables. The label or listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the wire or cable conforms to this requirement.

3.2.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his wire or cable conforms to the applicable requirements of the published standards including methods of tests of the Underwriters' Laboratories, Inc., Standard No. 133 for Varnished-Cloth Wires and Cables.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Unless otherwise specified, inspections by the supplier may be performed on a statistical or production lot basis to assure that the quality of manufacture is equal to the standards set by Underwriters' Laboratories, Inc. (see 6.2).

4.2 Sampling, nonmilitary.

4.2.1 *Lot.* Unless otherwise specified in the invitation for bids, the lot shall be formed as described in paragraph 3 of section 6 of J-C-98.

4.2.2 *Sampling for inspection and tests.* Unless otherwise specified in the invitation for bids, the sample shall be taken as described in paragraph 6 of section 6 of J-C-98.

4.3 Visual inspection, nonmilitary.

4.3.1 Each inspection unit in the sample, 4.2.2, as well as the package from which the inspection unit is taken, shall be examined visually before any tests are made. Visual inspection shall cover construction tinning of conductors, splices, marking, type of conductor, joints and repairs, and may be limited to that part of the package which is visible without unrolling.

4.4 General inspection, military.

4.4.1 *Definitions.* MIL-STD-109 applies for definitions of inspection terms used herein.

4.4.2 *Acceptance inspection.* Acceptance inspection shall consist of group A, group B, and group C inspection in 4.4.2.1 through 4.4.2.3.1. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from

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units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection or both.

4.4.2.1 *Group inspection.* Group A inspection including sampling shall be in accordance with table I and MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE I. *Group A inspection*

Inspection	AQL	
	Major	Minor
Visual and mechanical:	*Percent*	*Percent*
General construction Tinning Splices Marking Type of conductor (diameter) Joints Repairs	1.0	4.0
Electrical:		
Resistance of conductor Insulation resistance Dielectric strength	1.0**	*

* All electrical defects are considered major.
** 1.0 percent unless otherwise specified.

4.4.2.2 *Group B inspection.* This inspection including sampling shall conform to table II and to MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1.

4.4.2.2.1. *Group B sampling plans.* The group B sampling plans as listed in table II shall be as follows:

AQL	Inspection level for normal inspection	Inspection level for reduced inspection
Percent		
6.5	L-7	L-5

TABLE II. *Group B inspection*

Inspection	Sampling plan
Subgroup 1:	
Average thickness of insulation Minimum thickness of insulation (less than nominal 3/64 inch) Minimum thickness of insulation (nominal 3/64 inch and over) Thickness of braid Thickness of cotton tape Thickness of lead covering Thickness of varnished cloth Lay of pitch, lay of twist Tangent of angle between braid and axis of wire cable Tangent of angle between serving wrap and axis of wire or cable Picks per inch Yarn size Length of lay, wrap or serving Carriers and ends per carrier, braids	See 4.4.2.2.1
Subgroup 2:	6 specimens per 100,000 feet or fraction thereof.
Flame resistance, fibrous covering except tapes Dripping of compound	

4.4.2.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.4.2.2.3 *Procedure in the case of failure.* If an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

TABLE III. *Group C inspection*

	Sampling Plan
Tensile strength, varnished cloth Effect of heat, varnish coating	6 specimens per 500,000 feet

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4.4.2.3 *Group C inspection.* This inspection shall be as listed in table III.

4.4.2.3.1 *Specimens.* Six specimens shall be prepared and tested from each lot of 500,000 feet or fraction thereof of cable and wire. If each of the 6 specimens meet the specified requirements, the lot shall be accepted.

4.4.2.3.1.1 If 2 or more of the 6 specimens fail to meet the specified requirement, the lot shall be rejected.

4.4.2.3.1.2 If only 1 of the 6 specimens fails to meet the specified requirements, a second sample of 6 specimens shall be tested. Lot acceptance shall be permitted only if in the combined sample of 12 specimens not more than one specimen fails to meet the requirements. Failure of 2 or more specimens in the combined first and second samples shall result in lot rejection.

4.4.2.3.2 *Noncompliance.* If a lot fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and on all units of product which were manufactured with the same conditions, materials, processing, etc. If the Government inspector does not consider that the corrective action will enable the product to meet the special requirement or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3 *Disposition of nonconforming product.* Disposition of lots and sample units rejected during group A or B inspection shall be in accordance with MIL-STD-105. The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means to correct the fault:

(*a*) Sample units found defective during groups A, B, or C inspection.

(*b*) Inspection lots rejected during group A or B inspection.

(*c*) Product has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Tests.

4.5.1 *Nonmilitary.* The cable and wire shall be subjected to the tests set forth in Underwriters' Laboratories, Inc., Standard UL-133 to determine compliance with 3.1.

4.5.2 *Military.* The cable and wire shall be subject to the tests set forth in Underwriters' Laboratories, Inc. Standard UL-133 modified by the instructions in 4.4 to determine compliance with 3.1.

4.6 Compliance criteria, nonmilitary.

4.6.1 *Inspection.* If one or more of the packages, 4.2.2, fails to conform to the requirements for which inspection was made, 4.3, the lot from which the packages were taken shall be considered to have failed to meet the requirements of this specification.

4.6.2 *Tests.* Compliance of the lot with respect to the tests shall be based on the principles described in section 6 of J-C-98.

4.6.2.1 Any reel, coil, or roll of cable or wire which shows the presence of a defect or fails to meet the requirement when subjected to the electrical tests, shall be considered to have failed to meet the requirements of this specification. If after the defects are repaired and the roll, reel, or coil is again subjected to the dielectric strength and insulation resistance tests, the presence of a defect is not shown, the roll, reel, or coil shall be considered to have met the requirements of this specification for dielectric strength and insulation resistance.

4.7 Compliance criteria, military.

4.7.1 The compliance criteria shall be as set forth in 4.4.

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5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102).

5.1 Packaging.

5.1.1 *Levels A and B.*

5.1.1.1 *Coils (AWG sizes 14 to AWG 10).* Cable and wire shall be uniformly coiled and packaged 500 feet in a fiberboard box conforming to PPP-B-636, type I or type II, class 2, provided with the special knock-out feature normally used in commercial practice for dispensing purposes.

5.1.1.2 *Coils (AWG sizes 8 to AWG 2).* Cable and wire shall be uniformly coiled and packaged 500 feet per coil. The coils shall be spirally wrapped with waterproofed kraft paper conforming to UU-P-271, class E-2 and sealed.

5.1.1.3 *Reels.* Cable and wire shall be packaged on wood or steel reels suitably designed with strong flanges of sufficient diameter to protect the cable or wire from damage during shipment and storage. After the cable or wire has been wound on the reel, it shall be covered with a full wrap of waterproof kraft barrier material conforming to UU-P-271, class E-2. The wrap shall extend between the flanges and be secured in place with a minimum of two flat steel straps.

5.1.2 *Level C.* Cable and wire shall be packaged in coils or on reels of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1. *Coils (AWG sizes 14 to AWG 10).* Cable and wire packaged as specified in 5.1.1.1 shall be packed in nailed wood or wood cleated plywood boxes conforming to PPP-B-621 and PPP-B-601 respectively. Containers shall be fitted with waterproof case liners conforming to MIL-L-10547, class 2 and sealed in accordance with the appendix thereto.

5.2.1.2 *Coils (AWG sizes 8 to AWG 2) and reels.* Cable and wire packaged as specified in 5.1.1.2 and 5.1.1.3 shall require no further overpacking.

5.2.2 *Level B. coils and reels.* Cable and wire packaged as specified in 5.1.1.1, 5.1.1.2 or 5.1.1.3, shall require no further overpacking

5.2.3 *Level C.* The subject commodity shall be packaged and packed in substantial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2. Military agencies. In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Class, style, and size of cable and wire (1.2.1).
(c) Service voltage (1.2.2).

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(d) Number of conductors.
(e) Administrative provisions for inspection records (4.1).
(f) Lot formation, if other than specified (4.2.1).
(g) Methods of taking sample, if other than specified (4.2.2).
(h) Level of packaging and packing required (5.1 and 5.2).
(i) Special marking, if required (5.3).

6.2 If laboratory tests by the Government are required, purchasing officers should order wire or cable for test purposes in addition to the number of feet required for installation.

6.3 Underwriters' Laboratories, Inc. Standard No. 133 for Varnished-Cloth Wires and Cable permits a substitution of the spark test in place of part of dielectric strength and insulation resistance tests for 600 volt wire and cable. Since the spark test is usually conducted as a part of the manufacturing process, the dielectric strength and insulation resistance (when required) should be specified in the invitation for bids and arrangements should be made with the manufacturer for witnessing the tests in his plant.

6.4 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Rail:

Cable, Electric, Copper, Not Otherwise Indexed by Name.
Carload Minimum Weight 30,000 pounds.

Motor:

Cable, Electric, Copper, Not Otherwise Indexed.
Truckload Minimum Weight 30,000 pounds, Subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—Sig
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Air Force—Rome Air Development Center.

Copies of this specification may be purchased for cents each.

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